



Decentralised Identity Management solution for zero-trust multi-domain Computing Continuum frameworks

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ABSTRACT

The adoption of the Computing Continuum is characterised by the seamless integration of diverse computing environments and devices. In this dynamic landscape, sharing resources across the continuum is becoming a reality and security must move an step forward, specially in terms of authentication and authorisation for such a distributed and heterogeneous environments. The need for robust identity management is paramount and, in this regard, Decentralised Identity Management (DIM) emerges as a promising solution. It leverages decentralised technologies to secure and facilitate identity interactions across the Computing Continuum. Particularly, to enhance security and privacy, it would be desirable to apply the principles of Self-Sovereign Identity (SSI). In this paradigm, users have full ownership and control of their digital identities that empowers individuals to manage and share their identity data on a need-to-know basis. These mechanisms could contribute to improve security properties during continuum resource management operations. In this context, this paper presents the design, workflows and implementation of a solution that provides authentication/authorisation features to distributed zero-trust based infrastructures across the continuum, enhancing security in resource sharing and resource acquisition stages. To this aim, the solution relies on key aspects like decentralisation, interoperability, trust management and privacy-enhancing capabilities. The decentralisation leverages distributed ledger technologies, such as blockchain, to establish a decentralised identity ecosystem. The solution prioritises interoperability, enabling nodes to seamlessly access and share their identities across different domains and environments. Trustworthiness is at the core of DIM, and privacy is also considered, incorporating privacy-preserving techniques that individuals to selectively disclose identity attributes while safeguarding sensitive information. The implementation includes different operations for allowing continuum frameworks to be enhanced with decentralised authentication and authorisation features. The performance has been evaluated measuring the impact for the adoption of the solution. The most expensive task, the self-identity generation, takes only a few seconds (in our deployment) and it is only executed once. Authorisation tasks operate in the millisecond range, which is a totally invaluable time if incorporated into resource acquisition processes in frameworks such as Ligo, used in the scope of FLUIDOS project.

1. Introduction

The aim of establishing a seamless Computing Continuum is to have an overwhelming number of connected devices from the cloud to the edge, where any device plays an important role, from IoT devices to huge data centres. Achieving this Computing Continuum is a significant challenge as it requires the dynamism and flexibility of all aspects, including service deployment, resource optimisation, networking, privacy, and security. Although much importance is given to security and privacy, the reality is that it is often relegated to the background. In

this dynamic landscape distributed and heterogeneous full of devices, the effective management of digital identities assumes a progressively critical role, designed to manage and secure our identity information and to provide access to relevant services. This is particularly vital in ensuring security and privacy, especially concerning authorisation and authentication processes [1]. Traditional methods of identity management have proven to be insufficient, as centralised identity systems are no longer equipped to meet the demands of this decentralised and diverse landscape. Identities now need the ability to seamlessly

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navigate diverse boundaries, adjust according to context, and prioritise both privacy and security. This requires a paradigm shift towards identity solutions that are as adaptive, dynamic and privacy-centric as the environment in which they operate.

Decentralised Identity Management (DIM) revolutionises the concept of identity by shifting the centre of control from centralised authorities to the individuals themselves. At its core, DIM leverages decentralised technologies, notably distributed ledger systems like blockchain, to establish a robust, secure, and immutable infrastructure for managing identity attributes. DIM's reliance on decentralised technologies ensures that, access control schemes for authentication/authorisation, Decentralised Identifiers (DIDs) and transaction data are distributed across a network of nodes, reducing the risk of a single point of failure. Moreover, the distributed nature of the system ensures that the load is spread across various nodes increasing scalability. The utilisation of standards such as Verifiable Credentials (VCs) and DIDs plays a key role in enabling the seamless exchange of identity information. This not only promotes interoperability but also aligns with initiatives like data spaces in GAIA-X and the International Data Spaces Association (IDSA) [2], ensuring compatibility and connectivity across diverse and heterogeneous environments.

DIM approach also emphasises security and privacy measures, leveraging cryptography techniques such as privacy-preserving Attribute-Based Credentials (p-ABC) and Zero-Knowledge Proofs (ZK proofs). These advanced cryptography methods empower users with control over their identity information. It enables selective disclosure and ensuring that data is shared only as per the user's preferences and consent. Within the DIM framework, individuals embrace a concept known as self-sovereign identity. This grants them complete ownership and autonomy over their digital personas, endowing them with the authority to control and manage their identity attributes without requiring centralised authorities [3].

Relying on these identification systems together with other techniques such as trust-based and reliable systems [4], it is possible to offer a series of guarantees when performing continuum operations like remote resources acquisition. This new business model is fundamental in the current technological paradigm. It allows us to continue to face new service deployments, increase scalability, maintain quality of service, privacy and security. This paradigm is key in the way we can achieve a seamless Computing Continuum.

In line with these considerations, this work aims to solve the need for enhanced security in inter-domain collaborations, particularly in scenarios involving resource sharing within multi-domain computing continuum scenarios. Traditional authentication and authorisation mechanisms often struggle to provide adequate security in such environments due to their centralised nature, leaving sensitive data and resources vulnerable to breaches and unauthorised access. To address this challenge we provide the design, workflow, and implementation of a decentralised Identity Management (IdM)-based solution, tailored for multi-domain computing continuum frameworks. This solution can be seamlessly integrated into distributed Zero-Trust architectures, particularly those designed for the Computing Continuum. By providing a decentralised framework for managing identities, access control, and permissions across domains, our solution ensures heightened security while facilitating seamless resource sharing and collaboration. Thus, the main contributions of this work are the following:

- Design a zero-trust approach for managing authentication and authorisation in decentralised platforms of nodes working seamlessly in intra and inter-domain edge/continuum computation resource sharing. This innovative approach is aligned to the new continuum computing requirements such as enhanced security, scalability, real-time data processing, and the ability to dynamically allocate resources on demand across the continuum.

- Design, implement and evaluate in a realistic scenario those processes following SSI principles, based on DIDs and Verifiable Credentials enhanced with Zero-knowledge proofs, enabling flexibility and privacy features critical for such system. The SSI principles are particularly suitable due to their inherent ability to provide decentralised and user-centric identity management.
- Design the integration of the solution into a holistic multi-domain orchestration and resource management framework, like the FLUIDOS¹ project. We will use it as a reference for validating the solution's capabilities to support the Liquid Computing Continuum and resource acquisition. This contribution is expected to have a major impact on the different decentralised and distribution projects, providing security in authentication and authorisation processes while maintaining data privacy.

The rest of the paper is organised as follows: Section 2 provides the related work. Section 3 provides the solution overview. Section 4 shows the design and workflows for generating identities, join to an organisation as well as applying the solution during decentralised resource acquisition stages. Sections 5 and 6 provide implementation and performance results, respectively. Finally, Section 7 provides the conclusions.

2. Related work

Systems based on centralised access control have been the central core of security protocols of many domains and organisations. Along the years, the evolution of tools and mechanisms for centralised authentication and authorisation has been remarkable. Initially, systems relied on simple methods such as usernames and passwords stored in centralised databases. Over time, more sophisticated solutions emerged, such as LDAP (Lightweight Directory Access Protocol) [5] and Kerberos [6] protocols, enabling more efficient management of identities and access in corporate environments. The widespread adoption of online services led to the development of broader standards, such as OAuth [7] and OpenID Connect [8], allowing users to authenticate across multiple websites using their existing credentials. However, these systems still heavily rely on centralised authorisation servers to verify users' identities and grant access to protected resources. Just as the cloud is moving towards adopting a distributed approach in pursuit of Computing Continuum, access control systems are also benefiting from this approach to identity management. The user-centric perspective and Self-Sovereign Identity emerge, as explained by C. Allen in [9] and envisioned by the European Union with several initiatives like the European Self Sovereign Identity framework (eSSIF) [10].

In this regard, recent studies propose the utilisation of decentralised identifiers (DIDs) [11] and Verifiable Credentials (VCs) [12] as a highly distributed substitute for traditional authentication methods. N. Fotiou et al. [13] shows a proof of concept of a DID registry for authorisation of guests to interact with IoT devices using a distributed Policy Decision Point (PDP) and Policy Enforcement Point (PEP). A similar notion is used by J. Werner and C. Merkle [14], which complement it with a centralised identity provider following the OpenID Connect standard [15]. G. Peterson [16] presents a more dynamic PEP and PDP architecture enabling PEP and PDP interactions in a highly distributed manner using XACML.

The integration of access control and verifiable credentials with privacy-preserving technologies such as attribute-based credentials (p-ABC) helps achieving key privacy goals. These distributed environments are presented in Torres Moreno et al. [17], which shows the framework of technologies at a theoretical level. In [18], authors present the implementation and integration of Pointcheval-Sanders

¹ <https://www.fluidos.eu/>

Multi-Signatures (PS-MS) into a distributed identity management system. As part of the research work within the H2020 project ER-ATOSTHENES, J. García-Rodríguez et al. [19] propose a framework with support for distributed issuance (dp-ABC) for privacy-preserving attribute-based authentication and authorisation with VCs in IoT scenarios. On the other hand, Rajkumar and Ravi [20] presents an ABC-based pre-authorisation model for usage control with finite attribute domains. The authors show that their model with finite attribute domains has decidable safety even if arbitrary object creation is allowed. Subsequently, Rajkumar and Ravi [21] extended their work by developing a new model incorporating infinite attribute domains, also addressing decentralisation aspects.

The versatility of using DID can be seen in [22], where authors show a proof of concept of decentralised communication peer-to-peer, trustworthy, and privacy-preserving without the need for any external networking infrastructure. A great enabler for this technology is distributed ledger technology (DLT). In this regard, Ding et al. [23] shows a DLT-based architecture with the concept of SSaaS for customising SSI services. Blockchain is a type of DLT that has emerged as a multifaceted tool in the realm of data management and security as it has the capability to act as a registration authority, as proposed by A. Mühle [24], and as an authorisation system in distributed services, as shown in Yu L.'s research, specifically focused on mobile cloud services. Yu et al. [25] propose using smart contracts to enable real-time updating of access permissions.

Tan et al. [26] shows the use of blockchain technology as access control system (authorisation), along with the use of DIDs for the authentication of users and IoT devices in reducing access complexity. For its part, Rani et al. [27] makes use of blockchain and DIDs in the field of health to maintain the privacy and security of the data collected from patients. Even in other fields born from the rise of blockchain technologies such as the metaverse [28] where the DID credentials are used for avatar authentication within blockchain-based metaverse services, verification of avatars occurs within smart contracts, ensuring user privacy through zero-knowledge proof (ZKP). The study in [29] explores encryption techniques for secure management of Decentralised Identities (DID) in blockchain. It evaluates methods such as homomorphic encryption, hash function (MD5), and zero-knowledge proofs. It proposes a zero-knowledge proof based encryption approach to improve efficiency in DID data processing, considering the frequency and scope of data usage.

Related to decentralised zero-trust access control and identity management is the evaluation of participant's trustworthiness. In this field, DLTs are also currently identified a key enabler for auditability and decentralisation of the evaluation, as highlighted in [30]. The calculation of trust scores themselves is varied in the literature for different use cases, such as using a combination (weighted sum) of parameters like cyber risk or security level [31], indirect evaluation of events and interactions through reputation [32] or complex modelling based on behaviour analysis [33]. M. Bampatsikos et al. [34] propose a solution for the *cold-start* problem, prevalent in zero-trust scenarios, with an initial computation based on participant characteristics like its security features.

All in all, multiple technologies for identity and trust play a pivotal role in decentralised environments. Many areas within the Computing Continuum can benefit of this type of solutions, one of which is resource sharing. There are ongoing projects attempting to address inter-domain resource sharing from the cloud to the edge and the various challenges involved in achieving secure Computing Continuum. M. Ioro et al. [35] introduces a new term Liquid Computing that refers to an innovative approach proposed with the goal of creating a transparent continuum of resources and services on top of fragmented infrastructure. This concept has been used in the FLUIDOS project scope. Other similar projects aiming at pooling resources to cope with computational tasks at different levels of the continuum can be found in [36,37].

Unfortunately, accessing unknown resources on demand and in an automated manner is challenging, as is sharing data with "strangers". The apparent lack of integration in security aspects, authentication and authorisation in distributed architectures amplifies vulnerabilities in resource and data sharing. This paper sets out to address this issue and presents a comprehensive solution that can be effectively incorporated into distributed zero-trust architectures through decentralised identity management. This approach strengthens the foundation of trust in interactions within distributed systems, particularly on the (Liquid) Computing Continuum in intra and inter-domain scenarios.

Despite the various works in the literature gathered in this section, none cover the key aspects addressed in this paper. Most solutions rely on centralised identity management, undermining the privacy of the involved entities and creating issues of security such as single-point of failure, rigidity and excessive centralisation. This hampers the adoption of such solutions in the (Liquid) Computing Continuum. Some works such as clearly [38] manifest the need for veering from existing centralised solutions in such scenarios. However, it relies on a centralised controller to manage a distributed key system, not dealing with critical notions like privacy or authorisation processes. In other cases, authors do not provide detailed processes about how identity management is tackled nor specific aspects of a resource sharing scenarios [16]. Besides, privacy issues and dynamic identity and trust management are left as future work [25]. These are key issues for the multi-domain and Liquid Computing Continuum considered in this paper. On another note, many works deal with the technical aspects of decentralised identity management and its enabling technologies (c.f., e.g., [17,23,24,28,29]) or zero-trust based access control (c.f., e.g., [13,14,16]), which can fit with the studied scenario. However, they suffer from issues regarding privacy [13,14,16,26], lack of attribute-based authentication and authorisation [22,29,39], or, importantly, do not cover the specific needs of a trustworthy and dynamic Computing Continuum (e.g., [13,16,19,22,23,27,29]), an aspect that has been shown to be critical for the successful application of such identity and trust management solutions [19,27,28].

3. Solution overview

Fig. 1 shows the main modules and components over a decentralised environment that represents a computing continuum multi-domain distributed platform. In concrete, the solution is composed of a Trusted Governance Plane and a set of the domains that achieve form part of the continuum by providing/consuming local/remote resources. Whereas the solution could be potentially composed of many amount of domains and nodes, totally heterogeneous, the figure shows a simplification composed of two domains which in turn are composed of three different nodes.

3.1. Trusted governance plane

This plane is governed by trustable entities and provides governance of the platform catalog, trusted services, data and transactions stored in a *Verifiable Data Registry* (VDR). The *Catalog* manages data and transactions regarding the resources exposed and conditions provided by each domain. This means, each domain will decide what to expose and who will be able to use its resources. The VDR maintains a secure and tamper-proof record of data entries, and these entries can be verified independently. In our solution it registers participants, trusted issuers (i.e., trust anchors for identity data), public information as well as resource sharing policies provided by the domain's participant. Despite looking like a centralised entity, the VDR can be instantiated using blockchain technology. In fact, domain nodes could also deploy blockchain nodes for enhancing, for instance, scalability. Similar to the Catalog Services, the *Trust Services* will provide supporting processes required by the trust and identity management in the ecosystem such as management of the trust-related lists in the VDR. Note that, in

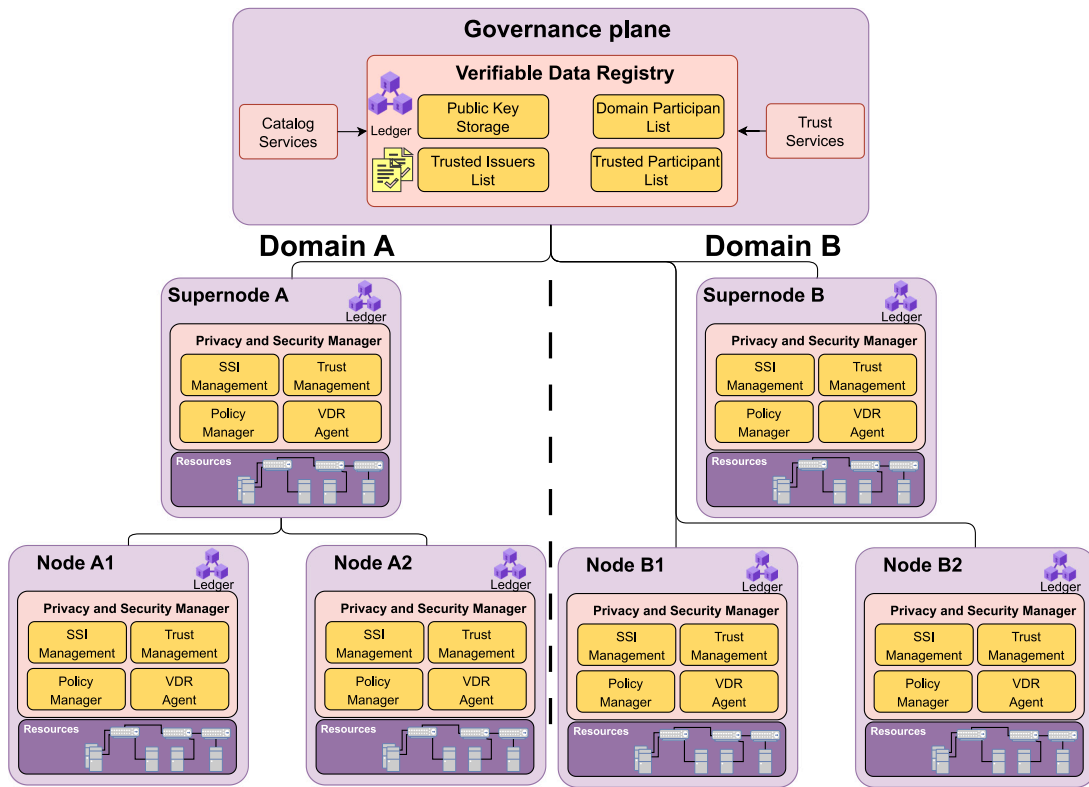


Fig. 1. Scenario.

both cases, the services may not be controlled by a central authority, but instead the group of entities that forms the ecosystem federation. What is more, their processes could be instantiated as smart contracts, leveraging the blockchain implementation of the VDR.

3.2. Domain nodes

A domain node represents a group of resources managed by the same control plane. For instance, a whole Kubernetes cluster could be defined as a domain node. Domain nodes can be considered regular nodes (e.g., Nodes A1, A2, B1, B2 in Fig. 1) or supernodes (e.g., Supernode A, Supernode B in Fig. 1) depending on if they provide special functions to other nodes like per-domain credentials generation or resources aggregation. As can be seen in the figure, the regular nodes from domain A are connected to the Supernode A. This is because the Supernode acts as an intermediary to manage credentials, unlike in domain B where regular nodes go directly to the trusted entity. The internal structure of a domain node can be divided into two different planes, the control plane and the resources plane.

3.2.1. Control plane

The Node control plane provides all the control and management modules required to administrate the resources plane. Although the specific components will depend on the nature of the continuum platform, roles such as orchestration, telemetry analytic, discovery and resource management are usually required to maintain the continuum. From these pivotal capabilities, our solution focuses on authentication and authorisation for the continuum, specially in terms of integrating with resource sharing. Thus, we concentrate on the modules required for providing distributed identity management that will enable said processes. In this context, we designed the Privacy and Security Manager component, composed of the SSI Management, Trust Management, Policy Manager and VDR Agent modules.

- **Self-Sovereign Identity (SSI) Management module** controls the identity of the Node generating its own identifier, acquiring and storing Verifiable Credentials and generating Verifiable Presentations (VP) from them to allow the identification of the node. Depending on the role of the node contribute on the identity of the other nodes inside the borders of its domain. For instance, acting as issuer, it could provide VCs to other nodes inside its domain. The module is also in charge of verifying the VPs and VCs presented from other nodes. Each node inside the domain will be able to verify credentials received from other nodes, inter or intra domain. It also publish in the VDR public cryptographic material in a way that is accessible and auditable at any time.
- **Trust and Reputation Manager module** is a critical component in establishing and managing trust relationships between entities in the system. The authorisation, as the authentication, must maintain the characteristic of decentralisation. In the realm of the Computing Continuum, the predominant approach is to adopt Zero Trust Architectures (ZTA). The zero trust paradigm is fundamentally based on the principle that trust is never implicitly assumed, but requires continuous evaluation. In this sense, the component will be key for ensuring interactions are only authorised after evaluation of identity and trust parameters at the Policy Decision Point. In this paper, we focus on how to address the actual authorisation process with privacy and security guarantees. Nonetheless, the complete scenario will also require the management of trust scores from various information gathered, such as attributes, behaviours or performance metrics. There are multiple works in the literature dealing with the aspects of trust score evaluation (c.f. Section 2). Among them, we find techniques that would fit with the design goals of this paper and the specifics of the application scenario, bringing advantages to the overall trust management. The use case naturally presents verified attributes for nodes such as organisation, resources and security measures (e.g., presence of Trusted Execution Environment) to apply computations to solve the cold-start problem [34].

Additionally, the authorisation process and compliance to the agreement for resource sharing are great candidates for the update of trust scores based on behaviour and reputation [32]. Lastly, as recently trending in the literature [30] the DLT can play a key role on the trust management framework for the ecosystem, offering verifiable and traceable exchange of information and even smart contracts for trust-related events.

- **Policy Manager** allows defining policies related to resource exposure and authorisation policies in terms of who will be able to access them, for instance in form of white/black lists. Thus, the module is needed for exposing the node features in the Catalog as well as for verifying resources acquisition requests.
- **Verifiable Data Registry (VDR) Agent** is a module within the solution that plays a role in managing and interacting with verifiable data. This is, the agent acts as an intermediary that facilitates the addition, retrieval, and verification of data within the registry.

3.2.2. Resources plane

The resources Plane refers to the layer of the node responsible for managing and utilising the physical and virtual resources available in a computing environment. For instance, in virtualised environments, the resources plane is the enforcement point for the creation, migration, and termination of virtual machines, ensuring efficient utilisation of physical hardware. In contrast, in containerised environments the resources plane oversees the deployment and scaling of containers. This plane also requires tools for monitoring the virtual/physical infrastructure in terms of the performance of computing resources, identifying bottlenecks and making adjustments as needed, as well as in terms of security according to the security policies established.

3.3. Solution techniques and technologies

In developing a decentralised authentication architecture, the use of decentralised identifiers (DIDs), verifiable credentials (VCs) and distributed ledgers technology (DLT) has been strategically integrated. This design reflects a paradigm shift in authentication methodologies, enhancing the security and privacy of the identity management process within this innovative collaborative framework. The incorporation of DID and VCs aligns perfectly with the concept of Self-Sovereign Identity, allowing participants to selectively control and present their identity. Together, these components contribute to a robust, decentralised authentication architecture that aligns with the autonomy and security inherent in the Computing Continuum scenario. Each of the technologies used is explained below.

3.3.1. Decentralised identifiers (DID)

The Decentralised Identifiers² (DIDs), as described in the W3C recommendation “Decentralised Identifiers (DIDs) v1.0” [11], represent a globally unique identifier revolution designed to allow individuals and organisations to generate their own identifiers in systems they trust. This unique feature makes DIDs particularly suitable for authentication architectures where individuals are responsible for managing their own identity. The distinguishing feature of DIDs lies in their design, which allows their holder full control over them without the need for third-party intervention. DIDs are essentially URIs (Uniform Resource Identifiers) that link a DID subject (in this case a node), to a DID document, thus facilitating secure and reliable interactions associated with that subject. Within each DID document, various elements can be articulated, such as cryptographic material, verification methods, services, or whatever is needed. Together, these components provide a robust set of mechanisms that enable a DID controller to convincingly demonstrate its control over the corresponding DID.

Fig. 2 shows an example of the DID format which is composed of three parts:

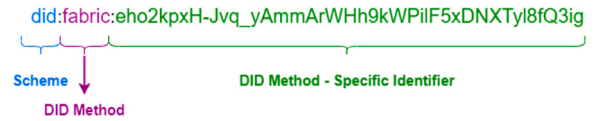


Fig. 2. DID Structure.

- **DID URI scheme:** It is denoted by the “did:” prefix. The scheme identifier indicates that the string following it is a Decentralised Identifier.
- **Identifier for the DID method:** Different DID methods exist, and each has its own way of generating and managing DIDs. This part specifies the method employed for creating the DID.
- **DID method-specific identifier:** This identifier is what distinguishes one DID from another within the specified method.

Adhering to the DID format specified, the generic DID scheme follows the URI scheme standards outlined in [RFC3986].

3.3.2. Verifiable credentials (VC)

Credentials play a key role in authenticating/authorising participants in an interaction. They will provide information that will enrich the identity information of the entity presenting it. The W3C recommendation ‘Verifiable Credentials Data Model v1.1’. [12] provides a model that aims to standardise this credential exchange. Using verifiable credentials their holders can generate verifiable presentations to share with verifiers to demonstrate that they possess identity attributes, particularly when enhanced with zero-knowledge proofs. The fast transmission capability of verifiable credentials and presentations together with the flexibility on the information they can represent further underscores their effectiveness. This makes them exceptionally convenient tools for establishing trust, even in remote scenarios such as distributed environments. It is important to emphasise that, in contrast to self-generated DIDs, a Verifiable Credential (VC) will generally not be validated by its holder (the node). Instead, the issuer is in charge of performing the validation process of the node/user/requestor. When a node seeks to obtain a VC, it must engage with an Issuer and present its DID for validation. Within the depicted scenario in Fig. 1, the Issuer could potentially be the Trust Services itself. Alternatively, the node has the option to refer to the “Trusted Issuer List”, a public registry containing trusted entities endowed with permissions to issue VCs. It is noteworthy, as illustrated in the scenario, that an Issuer might be restricted to generating VCs solely for its own domain. This could be the case when the domain A supernode is registered as issuer. It will be able to issue VCs only for its own domain.

3.3.3. Distributed ledgers technology (DLT)

The fundamental purpose of distributed ledgers, as essential components within the authentication architecture, is to provide an immutable and auditable registry for identity management and secure data exchange. The Distributed Ledger Technology (DLT) component enhances serves as a secure repository for identity-related data, such as public cryptographic material or Decentralised Identifiers (DID), enabling nodes to demonstrate ownership and control of their identities. Moreover, it also acts as a comprehensive registry for the traceability of shared resources, playing a crucial role in recording detailed information about the intended sharing of resources. This includes both successful transactions of resource exchange and those that did not materialise, providing a complete insight into the collaborative dynamics among nodes in the distributed environment.

4. Solution workflows

Following we describe the designed workflows for the enrolment and authorisation during the resource acquisition processes across the continuum.

² <https://www.w3.org/TR/did-core/>

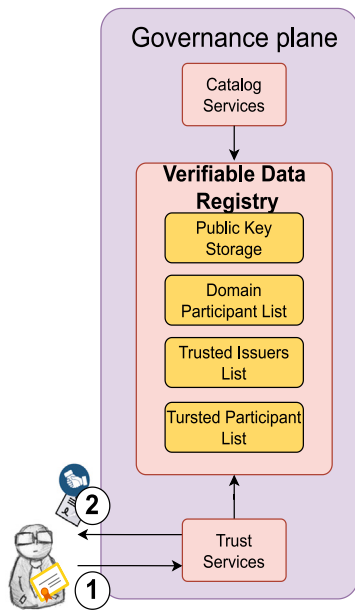


Fig. 3. Enrolment Process.

4.1. Enrolment process

In our solution, the process of incorporating domains/nodes into the Catalog with the intention of sharing/consuming resources is called Enrolment Process, and it is divided into two different phases. On the one hand, there is the enrolment agreement, which involves initiating the identification and verification of the domain seeking inclusion in the Catalog. On the other hand, the node enrolment phase occurs when nodes are instantiated, properly identified and their resources are exposed in the Catalog according to its resource exposition policies.

4.1.1. Decentralised enrolment agreement

The enrolment is the phase in which a domain representative starts the negotiation to be part of a collaborative catalog. During enrolment, the domain representative usually undergoes an establishment of access criteria or permissions. This could involve the evaluation of specific attributes, identities or compliance standards required for admission to the Catalog. This evaluation process, whatever automated or manual, forms the basis for determining the domain requesting inclusion. The access policy can be based on several strategies such as whitelists, blacklists or rules such as a firewall. If the entity passes the inclusion criteria, it will be included in the Catalog as a trusted domain in the Domain Participant List.

Fig. 3 shows a simplified workflow of the entity enrolment phase. The domain representative formally expresses its desire to be included in the Catalog. This implies to submit a certificate to the Trust Services, so they can verify that it is not fraudulent (Fig. 3 step 1). This process depends on the requirements of the Catalog. It could range from more restrictive to less restrictive (physical authentication to simple digital certificate of a business/person network). Once received, Trusted Services checks the validity of the certificate and also the access policies to see if the domain meets the requirements to be accepted into the Catalog. If the entity meets the criteria and is accepted, the Trust Services proceeds to request the addition of the public key of the domain certificate to the list of Domain Participant List in the Verifiable Data Registry (VDR) and informs the entity that it has been approved by creating a document that describes the prerequisites for participation in the Catalog (Fig. 3 step 2). These prerequisites may include shared resource commitments or other criteria. The formal agreement is established and the negotiation process is finalised. This

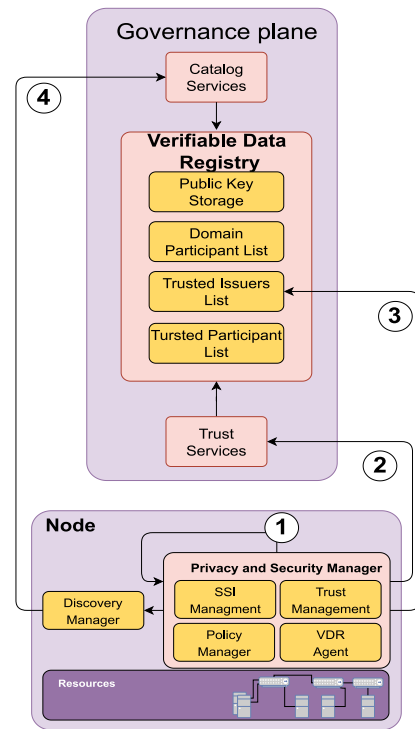


Fig. 4. Instance Process SuperNode.

consolidates the rights and responsibilities outlined in the formalised agreement.

This agreement and the inclusion of the domain representative in the Domain Participant list will allow further verification during the node enrolment phase. However, as we mentioned before, this process depends on the requirements of the Catalog and it could range from more restrictive to less restrictive. For instance, in a Zero trust approach, this process could be skipped so any node could enroll directly into the platform, where Trust Services would assign to them a low trust value by default until the node proves otherwise.

4.1.2. Decentralised node enrolment phase

The Node enrolment process is the next phase after a domain has successfully completed the enrolment agreement process and received approval. Fig. 4 shows the designed workflow that allows nodes generating its own identifiers as well as publishing the resources that are desired to be exposed into the Catalog.

The flow starts with the Privacy Security Manager module generating its own DID (Fig. 4 step 1), and sending the DID document to the governance plane (Fig. 4 step 2). The DID document is stored in the VDR and the DID is stored in the Participant List. For now, as the identity has not been verified at all, Trust Services could assign a low global trust score to it. Then, the node Privacy Security Manager module looks the Trusted Issuers List which is a public list of the issuers with the permission to generate Verifiable Credentials (VC) and requests to generate a VC. In case the node is a supernode, the request will also solicit to be a Trusted Issuer, so the supernode will be able to generate VCs for other nodes inside the domain. Independently if it is a supernode or a regular node, the node presents its DID along with certain attributes, including the domain to which it belongs to. Besides, the attribute that provides information regarding the domain is signed using the private key of the domain responsible, used during the enrolment agreement, thus, the Trusted Issuer Entity can verify the permissions associated with that DID, as well as updating its default trust score, and generating the VC. If the request solicited to be a trusted issuer, the node is added to the Trusted Issuer List with the

specific parameters (Fig. 4 step 3). The VC is provided to the node which stores it. Only the node stores its own VC. In the final phase, the Discovery Manager will use the DID for authenticating during providing the Catalog Services with both, the distinctive flavour and the latest information regarding the resources. This comprehensive data covers not only the technical specifications and attributes of the resources, but also includes relevant privacy details related to the shared resources. That is, the specific details of what information is intended to be shared and with whom, or who it is not intended to be shared with. For instance, an organisation node could desire sharing resources only with organisations that accomplish certain rules like being located in the same country or an specific overall trust score threshold. Listing 1 shows an example of authorisation policy modelled in Medium Security Policy Language (MSPL). We use this model as it extends previous research efforts and solutions, specially focused on security, such as the ones provided by I2NSF IETF group as well as some EU projects such as ANASTACIA-H2020 or INSPIRE-5GPlus. The language already provides multiple policy models for different capabilities and it is easily extensible.

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  <configuration xsi:type='RuleSetConfiguration'>
    <capability>
      <Name>AuthoriseAccess_resource</Name>
    </capability>
    <configurationRule>
      <configurationRuleAction
        xsi:type='authorisationAction'>
        <authorisationActionType>
          ALLOW
        </authorisationActionType>
      </configurationRuleAction>
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          </Attrs>
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        <authorisationTarget>
          <resourceID>ResourceSetA</resourceID>
        </authorisationTarget>
      </configurationCondition>
    </configurationRule>
  </configuration>
</ITResource>
```

Listing 1: Authorisation Policy Example.

In this example we use the *AuthoriseAccess_resource* capability, focused on providing Access Control to resources. Each capability extends specific actions and conditions. Actions and Conditions allow modelling specific set of actions (e.g. Allow/Deny) that must occur when specific conditions are accomplished. Those fields allow specifying access control requirements such as the concrete subject, its desired attributes and the target of the access control policy. For instance, only Domain B nodes located at Spain, with a trust score greater than 0.9 value will be considered. Regarding the authorisation target, the resource set identifier provided in the example refers to the resources that the organisation wants to share.

Finally, the information regarding shared resources is stored in the Catalog so it is available for the rest of participant domains that accomplish with the established constraints defined in the authorisation policy (Fig. 4 step 4). To provide an example of a supernode acting as a trusted issuer we provide the workflow shown in Fig. 5.

The Supernode, acting as a trusted issuer in the organisation, implies that regular nodes do not need to go through the process of requesting Trusted Services to get a VC. Instead, a regular node within the same domain of the scenario can directly request the VC to its supernode. If a regular node wishes to share its resources, it must generate its own DID and store it on the VDR, as the supernode did. Once this process is completed, the node requests the generation of a VC with the relevant permissions for a regular node (step 3. Fig. 5). Subsequently, it communicates to the Catalog the updated details of the exposed resources and access policies (step 4. Fig. 5), resembling the process carried out by the supernode.

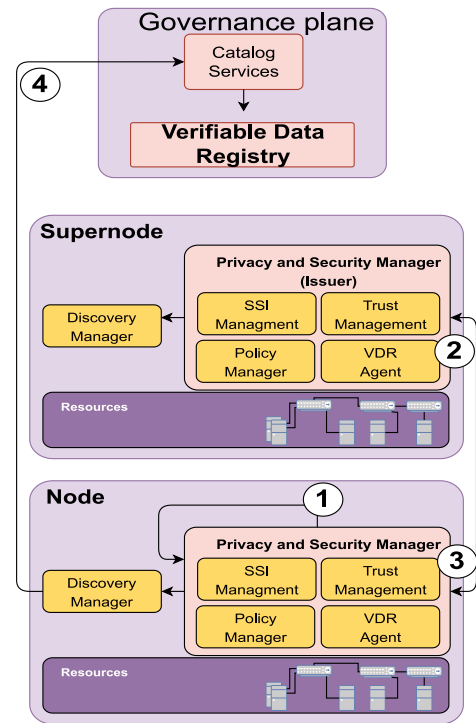


Fig. 5. Instance Process Node.

It is essential to highlight that whereas node enrolment occurs the first time the node has been instantiated, the resources update occurs each time the node's exposed resources or access policies are affected.

4.2. Decentralised authorisation during resource acquisition

In the dynamic distributed infrastructure landscape, spanning from the cloud to the edge, a sophisticated authentication and authorisation framework that aligns seamlessly with its decentralised architecture is required. Traditional centralised approaches, designed for a more static environment, are often inadequate for the fluid, decentralised nature of modern infrastructure deployments.

In the scope of our solution, Fig. 6 illustrates the process in which one domain (A) needs to acquire resources from another domain (B). To accomplish this acquisition, domain A will ask to the Catalog for external available resources. To do this, new components are introduced within the continuum framework. However, this paper does not go into the specific explanation and function of these components, since frameworks can vary in their components and invocation methods. Instead, this paper and specifically this section is more focused on how AuthN/AuthZ is applied in a use case where a domain needs to acquire resources. These components are: *Resource Acquisition Manager*, in charge of leading the negotiation process to obtain or share resources. In turn, the *Resource Importer/Exporter* module manages resource requests and deliveries, acting as an intermediary in the acquisition and sharing process. Finally, the *Contract Manager* is responsible for recording negotiations as transactions in the DLT, thus formalising the agreement reached. In addition, essential databases such as Available Resources and Peering Candidates support this resource sharing flow. As each node, during enrolment stage provided the exposed resources and the access policies, the Catalog only provides those that match according to the DID provided in the request. The domain A will select the most suitable external resources candidate, according to its own policies. In this example, domain B will be selected. Domain A will generate a Verifiable Presentation (VP) specific for domain B, providing specific fields that could be required by B. Domain B gives domain

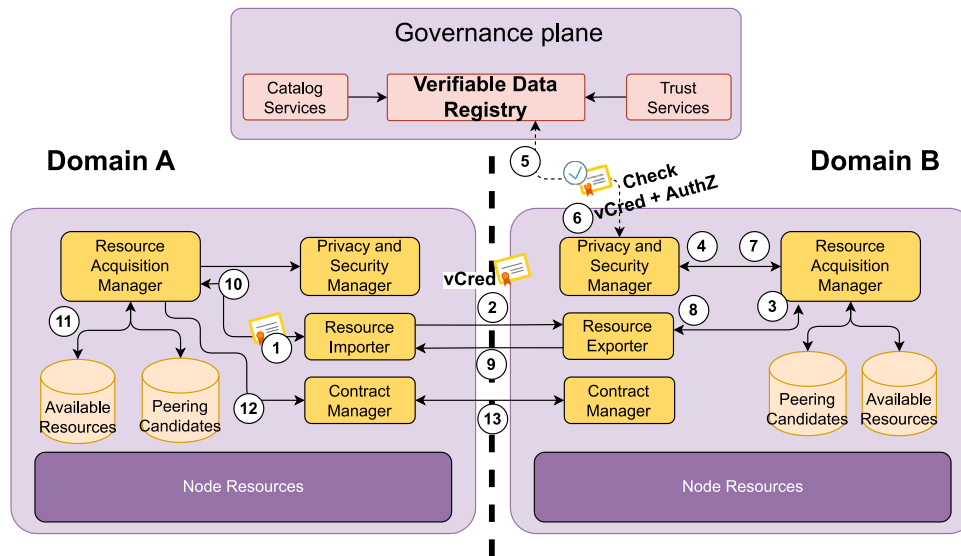


Fig. 6. Workflow AuthN/AuthZ in Resource Acquisition.

A authorisation, relying on the available information in the VDR, to acquire part of its resources because its policies so determine. In this flow appear two important conceptual roles that we would like to highlight:

- **Policy Decision Point (PDP):** The Policy Decision Point is a component responsible for evaluating access requests against predefined policies and generating any session-specific authentication token or credential used by a client to access a resource. The PDP typically communicates with the PEP (Policy Enforcement Point) to provide decisions, which are then enforced by the PEP. In this case, the role of the PDP is taken on by the Privacy and Security Manager.
- **Policy Enforcement Point (PEP):** The Policy Enforcement Point is a component in an access control system responsible for enforcing access decisions made by the PDP. The PEP is involved in enabling, monitoring, and potentially terminating connections between a subject (entity seeking access) and a resource based on the decisions received from the PDP. In this case, the role of the PEP is taken on by the Resource Acquisition Manager.

Following, the decentralised authorisation during resource acquisition process is explained in detail. The flow starts with the Resource Acquisition Manager invoking the Resource Importer (Fig. 6 Step 1.). The request includes the Verifiable Presentation, generated on demand by the Privacy and Security Manager as it plays the PDP role. The VP was custom generated to be compliant with the policy of the remote node candidate in the domain B. For instance, the domain B could require specific attributes in the VP. The Resource Exporter in domain B receives the VP together with the resource acquisition request that domain A is asking for (Fig. 6 step 2). It forwards the request to the Resource Acquisition Manager (Fig. 6 step 3) which needs to verify if the resource acquisition must be granted or not. To this aim, it asks to Privacy and Security Manager for an answer, as the Privacy and Security Manager plays the PDP role. (Fig. 6 step 4.). The Privacy and Security Manager queries the VDR for the public key associated to that VP and the defined credential schemes (Fig. 6 step 5 and 6), the public key can be retrieved due to the VP contains the DID, and inside the DID document stored in the DLT is the public key. After validating the authorisation, if everything is correct, and the The Privacy and Security Manager of the node in domain B grants approval according to its policies to allow the allocation of resources. Then, the Resource Acquisition Manager, acting as the Policy Enforcement Point (PEP), permits access to its resources as dictated by the PDP. At this point, an

access token could be generated for providing access between clusters at technical level. This token could be provided as part of the success notification (Fig. 6 step 7 and 8).

The Resource Acquisition Manager in domain A node is then notified about the request has been accepted (Fig. 6 step 10) so it adds the acquired resources to the database of Available Resources (Fig. 6 step 11) and finally it invokes the Contract Manager (Fig. 6 step 12) to generate a favourable resource acquisition transaction between domain A and domain B, securely stored in the VDR (Fig. 6 step 13).

5. Implementation

This section delves into the implementation of the decentralised authentication/authorisation solution that will be validated in the scope of the FLUIDOS project, which addresses the challenges of the Computing Continuum. To create our decentralised, immutable, secure, interoperable and traceable overlay we leverage the Hyperledger Aries Agent,³ seamlessly integrated with the Hyperledger Fabric.⁴ On the one hand, Aries is a set of tools and libraries designed to provide a shared, reusable, interoperable infrastructure for decentralised identity and blockchain-interoperable projects. It focuses on enabling the exchange of verifiable credentials and the establishment of decentralised identity systems. On the other hand, Fabric is a permissioned blockchain platform for developing enterprise-grade applications. It provides a modular architecture with a pluggable consensus mechanism, allowing organisations to tailor the network to their specific requirements. Fabric is designed for use cases such as supply chain, finance, and healthcare. To further elevate the desirable capabilities, the p-ABC implementation detailed in [19] was incorporated to the solution. It provides cutting-edge cryptography that enables the creation of derived verifiable presentations and zero-knowledge proofs. The synergy of these components has converged to form a powerful interface, empowering the solution with the indispensable capabilities needed for seamless identity management within the designated scenario.

To seamlessly integrate Hyperledgers, enabling communication between Hyperledger Aries and Fabric, using Fabric as a Verifiable Data Record (VDR), and providing functionalities such as DID generation, enrolment, authentication and authorisation we implemented the following modules:

³ <https://github.com/hyperledger/aries-framework-go>

⁴ <https://www.hyperledger.org/projects/fabric>

- **DID Management smart contracts:** To enable the use of Hyperledger Fabric as the VDR for the identity data, we implemented smart contracts for registering and resolving DID.
- **VDR Agent:** This component is a specialised VDR client that leverages the network of Hyperledger Fabric nodes. In practical terms, nodes use this module to expose their DIDs publicly and resolve DIDs within the ledger to retrieve the corresponding documents through calls to the appropriate smart contracts.
- **SSI Management:** This component defines methods leveraging the Aries agent's capabilities to address identity management requirements outlined in sections 3–4. Within this component we can find the following implemented processes:

- **Generate DID:** Create DID with keys/purposes as specified in request (c.f. Listing 2), and register it in the DLT. Any entity in the scenario can thus self-manage its identification data, and particularly its keys.
- **Enrolment:** Do an enrolment process against an issuer, obtaining a new VC. With this method, a node will be able to obtain a verifiable credential that asserts its identity attributes for its later use during authentication and authorisation processes.
- **Generate Verifiable Presentation:** Generate a Verifiable Presentation of a VC for an authorisation process in accordance with a policy that allows selective disclosure of attributes and with which the credential holder will only reveal the attributes needed in each authorisation.
- **Credential/Presentation verification:** Verify a Verifiable Credential or a Verifiable Presentation, returning a boolean of the verification result to complete an authorisation process.

Following, we provide detailed information about the SSI Management processes implementation, which have been developed as different endpoints part of a REST HTTP API in the Golang Framework.

5.1. Generate DID

The purpose of this method is to enable an entity to create a DID that will be used as its identification in the system. During this process, the node creates a DID with its DID document associated with the specification dictated in DID W3C standard [11]. Once the DID document is created, it is published into a DID registry. Particularly, we establish a DID Method identified by the string *fabric* that will handle creation and resolution operations through our custom VDR-Agent implementation. This will result in the publication and resolution of the DID into a permissioned blockchain implemented with Fabric through smart contracts. The contents of the DID Document will be a series of public keys associated to standard purposes (authentication, assertion...) created within this process and stored in the entity's wallet.

Fig. 7 shows the activity diagram for the *Generate DID* operation. The first step after checking the input parameters is to open the entity's private wallet. Using the wallet, a pair of keys is created for each type requested, public key along with metadata like id and purpose is added to the *verificationMethod* field of the DID Document, while the private key is kept in the wallet. Once we have filled the DID document information, a request is made through the VDR *createDID* interface which will finalise the creation of the DID and register it in the VDR according to the *fabric* method.

Listings 2 and 3 show an example of a request to this method and the resulting output, as used in the evaluation section. The request includes a key whose purpose will be the authentication of the DID's controller, i.e., the entity executing the method. The second key will be used for assertions, that is, signing documents. Particularly, it will be a key associated to the p-ABC cryptography for issuing verifiable credentials of the specific type managed by the entity, which will have four identity attributes. We will also obtain a name as a parameter, which will be useful for simpler reuse of the generated material.

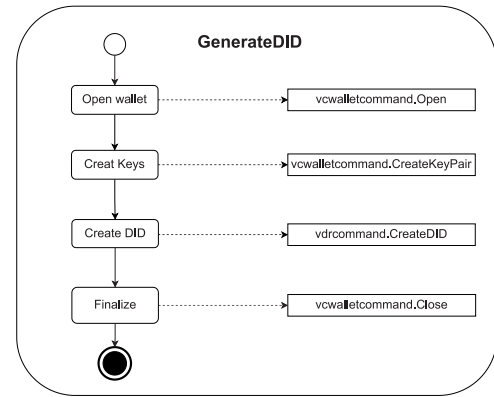


Fig. 7. Generate DID Operation.

```

{
  "keys": [
    {
      "keyType": {
        "keytype": "Ed25519VerificationKey2018"
      },
      "purpose": "Authentication"
    },
    {
      "keyType": {
        "keytype": "Bls12381G1Key2022",
        "attrs": [ "4" ]
      },
      "purpose": "AssertionMethod"
    }
  ],
  "name": "SuperNode"
}

```

Listing 2: Authentication and Assertion purpose DID Request example.

```

{
  "didDoc": {
    "@context": [
      "https://www.w3.org/ns/did/v1"
    ],
    "id": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27Wm4waHzs16MK6rSI",
    "verificationMethod": [
      {
        "controller": "did:fabric:GlqD3rBo5m_1vGCB4Jule...",
        "id": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27...",
        "publicKeyBase58": "2msepAk7gZqEvCsfoYFcqgVFI...",
        "type": "Ed25519VerificationKey2018"
      },
      {
        "controller": "did:fabric:GlqD3rBo5m_1vGCB4Jule...",
        "id": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27W...",
        "publicKeyBase58": "1xgswXrljZW5AQnBZZfZcieStMcub...",
        "type": "Bls12381G1Key2022"
      }
    ],
    "authentication": [
      "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE2..."
    ],
    "assertionMethod": [
      "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE2..."
    ]
  },
  "created": "2023-11-29T11:32:53.395418047Z",
  "updated": "2023-11-29T11:32:53.395418047Z"
}

```

Listing 3: Issuer DID example.

5.2. Enrolment

The enrolment process will allow an entity to obtain a Verifiable Credential from an issuer, which will be later used in authentication and authorisation processes. Particularly, in the scenario, a node will be able to request a credential over its DID and attributes from an issuer from the Trusted Issuer List. To ensure the authenticity of the information, the request will contain identity proofs, such as a signature carried out by the domain's responsible, whose certificate was added in the system during the domain enrolment process. In the implemented validation scenario, apart from the mandatory *organisation* attribute,

the node will present its role in the organisation, the physical address of the machine the node is hosted in and a name with the respective proofs, as seen in Listing 4. The issuer then checks the signature of the request attributes and agrees to generate a signed verifiable credential (c.f., 6), which the node stores into its wallet, completing the process.

```
{
  "url": "https://issuer:9082",
  "theirDID": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27Wm4waHzs16MK6rSI",
  "idProofs": [
    {
      "attrName": "holderName",
      "attrValue": "HolderNodeUMU",
      "proof": "HvPAAd123kcVc39317mb..."
    },
    {
      "attrName": "holderRole",
      "attrValue": "FluidosNode",
      "proof": "Iut21vxAjOnaHJ..."
    },
    {
      "attrName": "organization",
      "attrValue": "UMU",
      "proof": "2as312aADMI62HPvgxey..."
    },
    {
      "attrName": "physicalAddress",
      "attrValue": "50:80:61:82:ab:c9",
      "proof": "FjksUJx0DMI62HPv..."
    }
  ]
}
```

Listing 4: DoEnrolment request example.

To initiate the process, the node will call the *DoEnrolment* method (c.f. Fig. 8). The parameters used in the implemented scenario are shown in Listing 4. Apart from the aforementioned attributes and proofs, it includes the IP address and the DID of the issuer, which were collected from the Trusted Issuer List. The DID serves as assurance of the authenticity of the issuer. The attributes included in the request correspond to those defined in the example credential type, in this case a type issued by a supernode in the organisation for its domain. In Listing 5, we include the example context that enables the use of JSON-LD with said credential type and attributes, as reflected in W3C's standard [12].

```
{
  "@context": [
    {
      "@version": 1.1
    },
    "https://www.w3.org/ns/odrl.jsonld",
    {
      "ex": "https://inf.um.es/vcreds/",
      "fluid": "https://fluidos/vcreds/",
      "schema": "https://schema.org/",
      "InfUMCredential": "ex:InfUMCredential",
      "nodeName": "ex:nodeName",
      "nodeRole": "ex:nodeRole",
      "organization": "fluid:organization",
      "physicalAddress": "ex:physicalAddress"
    }
  ]
}
```

Listing 5: Example of Credential Template JSONLD.

After preparing the inputs, the node's method will call the issuer through the **AcceptEnrolment** endpoint (c.f. Fig. 8). Here, the issuer will parse the information received, verifying the proofs associated to each attribute. This process is performed by a list of validators, from which the correct validator will be selected according to the proof type, allowing the extensibility of the process to other proving mechanisms. In the implemented validation scenario, the proof is a signature that can be verified with respect to the organisation's (UMU) responsible certificate. If the validation is successful, the method will build the credential from the attributes along with metadata like the issuance date, and through the **Issue** interface generate the final Verifiable Credential signed with the issuer's private key. Lastly, this credential will be returned to the node, which will verify it has been correctly generated and store it for future use (**StoreCredWallet**).

The credential resulting from this process in the validation scenario is shown in Listing 6.

```
{
  "credential": {
    "@context": [
      "https://www.w3.org/2018/credentials/v1",
      "https://www.w3.org/2018/credentials/examples/v1",
      "https://ssiproject.inf.um.es/security/psms/v1",
      "https://ssiproject.inf.um.es/poc/context/v1"
    ],
    "credentialSubject": {
      "holderName": "HolderNodeUMU",
      "holderRole": "FluidosNode",
      "organization": "UMU",
      "physicalAddress": "50:80:61:82:ab:c9"
    },
    "expirationDate": "2023-11-30T15:19:38.07358607Z",
    "id": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27Wm...",
    "issuanceDate": "2023-11-29T11:32:58.07358607Z",
    "issuer": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27Wm4...",
    "proof": {
      "created": "2023-11-29T11:32:58.129925741Z",
      "proofPurpose": "assertionMethod",
      "proofValue": "BAfE3ocybW1Rumz6AhsUxifzop1tA7M...",
      "type": "PsmBlsSignature2022",
      "verificationMethod": "did:fabric:GlqD3rBo5m_1vGCB..."
    },
    "type": [
      "VerifiableCredential",
      "InfUMCredential"
    ],
    "credStorageId": "did:fabric:GlqD3rBo5m_1vGCB..."
  }
}
```

Listing 6: Verifiable Credential example (doEnrolment result).

5.3. Generate verifiable presentation

```
{
  "@context": [
    "https://www.w3.org/2018/credentials/v1"
  ],
  "type": "VerifiablePresentation",
  "verifiableCredential": [
    {
      "@context": [
        "https://www.w3.org/2018/credentials/v1",
        "https://ssiproject.inf.um.es/security/psms/v1",
        "https://ssiproject.inf.um.es/context/exampleContext/v1",
        "https://w3id.org/security/bbs/v1"
      ],
      "credentialSubject": {
        "organization": "UMU"
      },
      "expirationDate": "2023-11-30T15:19:38.07358607Z",
      "issuanceDate": "2023-11-29T11:32:58.07358607Z",
      "issuer": "did:fabric:GlqD3rBo5m_1vGCB4JuleBiRE27Wm4...",
      "proof": {
        "created": "2023-11-29T16:21:06.159613628+01:00",
        "nonce": "bm9uY2VGb3JaalByb29m",
        "proofPurpose": "assertionMethod",
        "proofValue": "AAIRBAQXu54...",
        "type": "PsmBlsSignatureProof2022",
        "verificationMethod": "did:fabric:GlqD3rBo5m_1vGCB..."
      },
      "type": [
        "InfUMCredential",
        "VerifiableCredential"
      ]
    },
    ...
  ]
}
```

Listing 7: Generate VP example result.

This process is used to generate a Verifiable Presentation from the stored identity credential(s) according to the needs of a specific authorisation process, as reflected in the respective policy. The generated presentation will not contain the original credential, but instead a derived version secured with zero-knowledge proofs so that only the specific information requested in the policy is included. For instance, in the evaluation scenario, the verifier – that is, the node that receives the petition for computing resources – is interested in learning the organisation of the solicitor node, as it is only interested in sharing resources with organisations from a specific region or sector. However, the physical address, role or name, which may be used by the solicitor for management flows inside its own organisation, are not relevant to the authorisation flow. In fact, sharing them would result in a privacy breach and potentially a security issue. The result of applying the process in the validation scenario can be found in Listing 7. Note that the proof in the credential is now a zero-knowledge proof, but it can still be checked only against the public key of the original issuer, keeping the chain of trust intact.

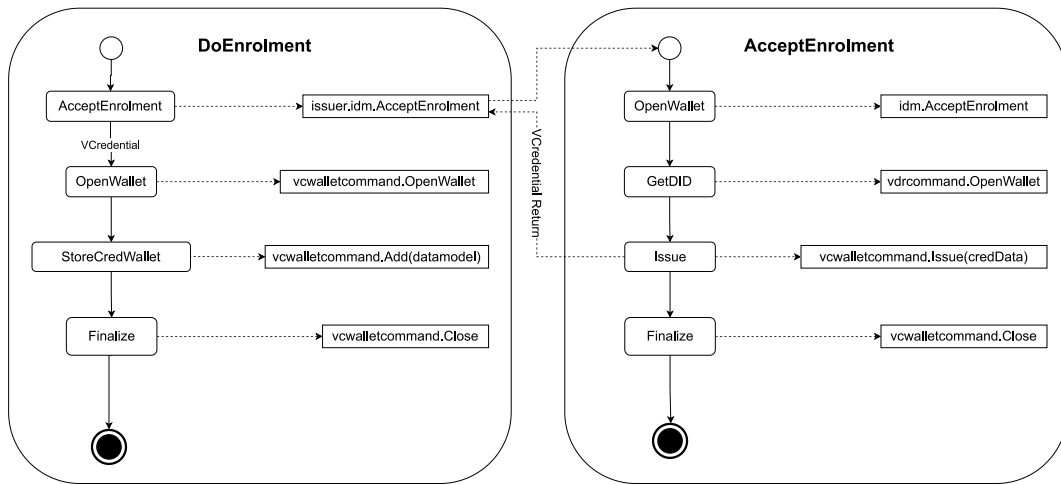


Fig. 8. DoEnrolment + AcceptEnrolment activity diagram.

5.4. Verify

This method has been created to give a relying party the ability to verify that a credential or presentation is signed correctly to complete an authorisation process. In the validation scenario, that means that the node that received the request for resources can check that the verifiable presentation that was sent to it, revealing only the solicitor's organisation, is valid. To do so, it will require the resolution of the DID of the issuer, to get the respective public key and perform the cryptographic operation. Once the validity check is performed, the relying party can be assured of the authenticity of the claim, and act according to its policies regarding resource sharing.

5.5. Extensibility and interoperability

We have presented the current implementation tailored to the specifics of the paper's application case. However, its interoperability and extensibility have been key design goals. This has been reflected on the use of W3C Verifiable Credentials as the vector for the privacy-preserving authentication based on p-ABC. This allows seamless integration with other digital credentials. Particularly, it enables both the re-use of the models and implementation in other contexts where SSI is attractive, such as personal identity wallets. Conversely, as long as compatible signature suites are supported, the proposed model could be fulfilled with heterogeneous SSI agent implementations.⁵ In a similar vein, the overall approach is aligned with current initiatives like European data spaces such as proposed by GAIA-X and IDSA. Particularly, each participant having PDP and PEP components for zero-trust authorisation, enabled by SSI and federated trust anchors is inline with the convergence of the Data Spaces Business Alliance.⁶ Thus, the proposed models and assets can enable and take advantage of further development in this scope. Further, this highlights the potential of achieving a collaborative and interconnected ecosystem, homogeneously covering the diverse security, trust and privacy needs of data spaces and the Liquid Computing Continuum.

6. Performance evaluation

This section provides a detailed performance evaluation of the solution. Fig. 9 shows the testbed we deployed in a realistic environment with the aim of showcasing the applicability and operational efficiency

of the proposal. It is composed of two different domains. One supernode and a regular node were deployed in the University of Murcia (UMU) whereas other supernode was deployed at University Center of Defense (CUD) at the Spanish Air Force Academy. The governance plane, composed of a fabric test network which acts as a VDR was also deployed at UMU, however, it could be deployed at any other trustable location at the cloud. Each node implements SSI Management, PDP, and VDR Agent, components form the Privacy And Security Manager as our emphasis is on assessing the efficiency of the decentralised identity-based authentication/authorisation process by measuring the time required for each method's execution from those exposed in the previous section.

For evaluating the performance, we measured the completion of the enrolment process of a UMU node against the UMU supernode, as well as the use of the generated credential for an authentication and authorisation process for resource acquisition at CUD. The experiments have been executed in a batch of 50 executions for each operation. Each node has been deployed as a single machine with 4 CPU and 32 GB of RAM, while the proof of concept blockchain has been deployed using OpenStack. Each Fabric virtual host consists of 2 GB of RAM and a single virtual core. The Fabric infrastructure consists of 2 organisations with one peer each, a certification authority and an Orderer node. Every Hyperledger machine is running Docker v19.

We measure the time the set-up process takes since the UMU node and supernode start the DID generation until the resulting document is ready. Fig. 10 shows the results obtained, where our solution takes around two seconds to generate a DID. Also, there is no significant difference when it comes to a supernode that can issue credentials or when it is a regular node that cannot, as the cryptographic key generation process is very cheap [18]. The task as a whole is the most expensive due to the high-cost of communicating with the blockchain and the execution of a writing smart contract. This could potentially be reduced with the use of alternative consensus algorithms like Proof of Stake, in contrast to the default Proof of Work used in Hyperledger Fabric. Nonetheless, the process of generating DID's is only done once, when the node is instantiated, so the result is not highly critical to the applicability of the solution.

In the following we provide measurements on key aspects of the scenario: the node enrolment time, VP generation and credential verification for resource acquisition. For the node enrolment, we measured the time since the UMU node starts the enrolment process after obtaining a DID until the credential is received and stored (i.e., **DoEnrolment** and **AcceptEnrolment** processes, as shown in Section 5.2). *GenerateVP* measures the time since the UMU node SSI Manager receives the presentation generation request until it is returned. Finally *Verify* measures the time required for verifying the generated verifiable presentation at

⁵ See, e.g., <https://w3id.org/>

⁶ <https://data-spaces-business-alliance.eu/>

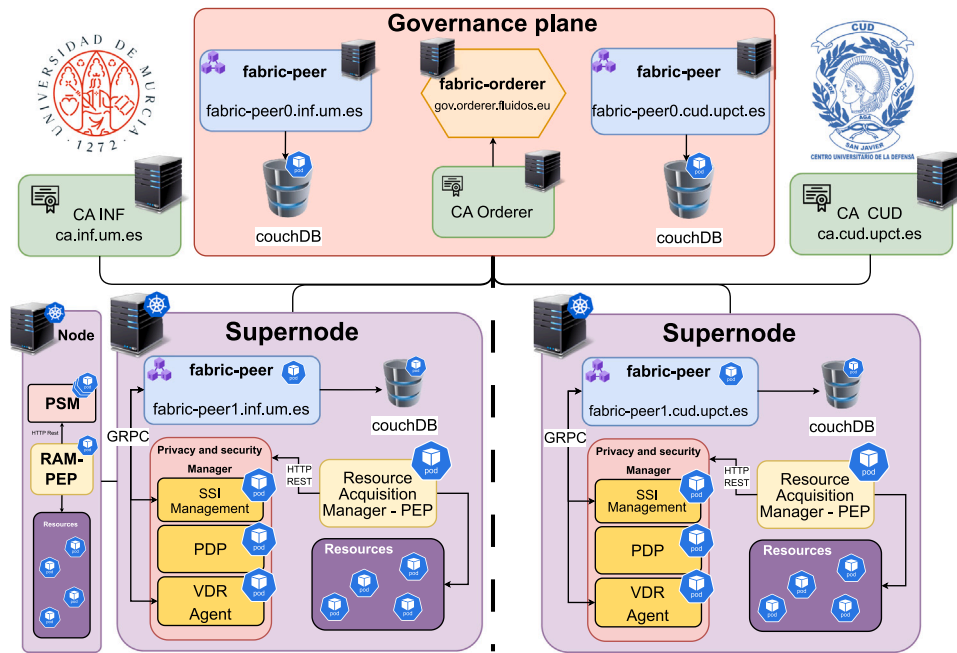


Fig. 9. Multi-domain testbed.

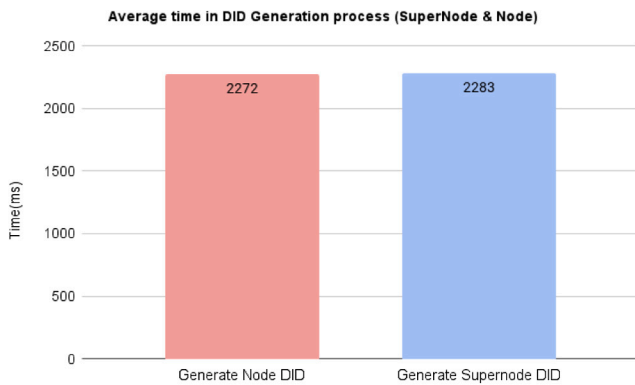


Fig. 10. Generate DID average time.

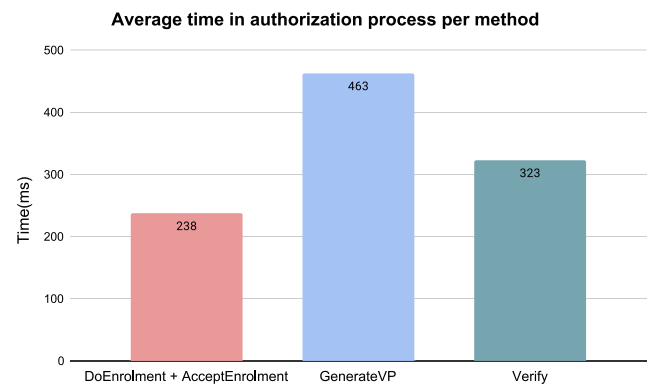


Fig. 11. Average time in authorisation process per method.

CUD supernode SSI Manager with the aim of granting/denying access to resource acquisition.

Fig. 11 shows that the operations are much faster than DID generation. Enrolment takes about 238 ms, VP generation takes about 463 ms and the verification also takes about 323 ms. The difference between them is due to the process of signing being significantly faster than the associated zero-knowledge proof [18]. An interesting observation is that VP generation is slower than verification, while in terms of cryptography the ZK proof generation is a bit faster than its verification. In this case, the heavy processing needed for managing the wallet credentials and JSON files during the VP generation is the biggest factor in the execution time. This is consistent with the results in [19].

In this case, we observe that the execution of all the attribute-based identity related processes have an average time of 1024 ms. However, only the last two steps (Generate VP and Verify) need to happen in all authentication and authorisation processes, and they take an average execution time of 786 ms. Considering that our solution operates in the millisecond range, and contrasting this with existing processes, such as resource acquisition in the Ligo [35] framework used in the FLUIDOS project, which takes about 2.5 s in the best case scenario and 4 s in the worst case, not including authorisation processes. In this context, we observe that our solution represents only 31% of

the time in the best case and 19% in the worst case. Therefore, the integration of our solution in this process does not introduce critical or appreciable delays, but also gives added value in terms of security and privacy. In the end, it has been demonstrated that frameworks such as Ligo can seamlessly incorporate decentralised authentication and authorisation, such as the one presented in this paper, without concern for performance impact.

Fig. 12 shows the percentage distribution for the decentralised authorisation process, where we can see that the bulk of the time in this process is spent on the generation of verifiable presentations and verification, with the issuance of credentials being the lowest in time, a little higher than the half of the generation of verifiable presentations and a little lower than the verify process.

Additionally, we measure the size of the credential and verifiable presentation exchanged during the enrolment and authorisation process. This reflects the communication overhead of using the proposed system based on SSI. Fig. 13 shows the average message size measured in the experiment. There is a clear difference between the overhead during enrolment due to the credential, and the overhead during authorisation due to the verifiable presentation. This is expected as the presentation includes an additional signature by the node, and more importantly a zero-knowledge proof over the identity attributes, which

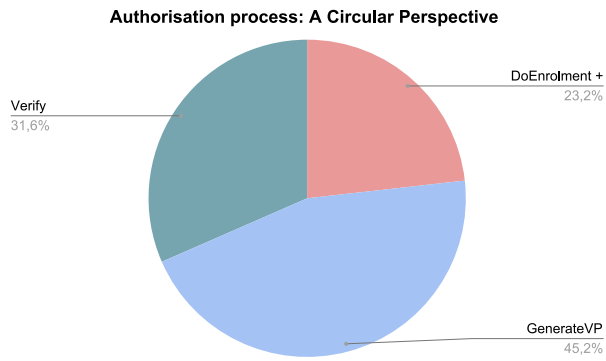


Fig. 12. Authorisation process.

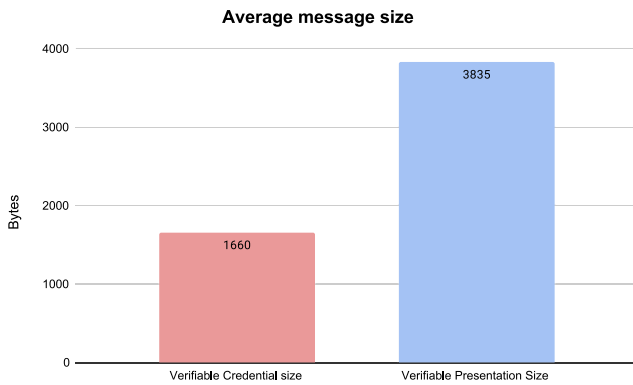


Fig. 13. Average message size.

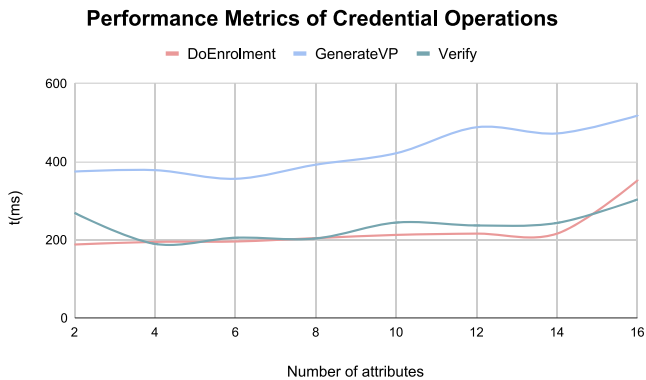


Fig. 14. Metrics of credential operations with varying number of attributes.

is larger in size than the original signature [18]. In any case, the overhead is under 4 kilobytes, which is very reduced for our use case.

On another note, the complexity of these operations changes depending on the number of attributes that are added to a credential and that are revealed or hidden in a presentation. Thus, we measured the impact of the number of attributes on the measurements and potential scalability. This is, how the time of an authorisation process increases depending on the attributes that are used to generate the credentials and verifiable presentations (ZK Proofs). To this aim, we progressively increased the number of attributes in our tests. We considered cases with 2, 4, 6, 8, 10, 12, 14, 16 attributes per test, revealing $n - 1$ attributes. Each case was repeated 50 times for obtaining consistent measurements. For this part of the experiment we do not consider registration of DIDs since this process is only carried out once for each node and will be unaffected by these variables.

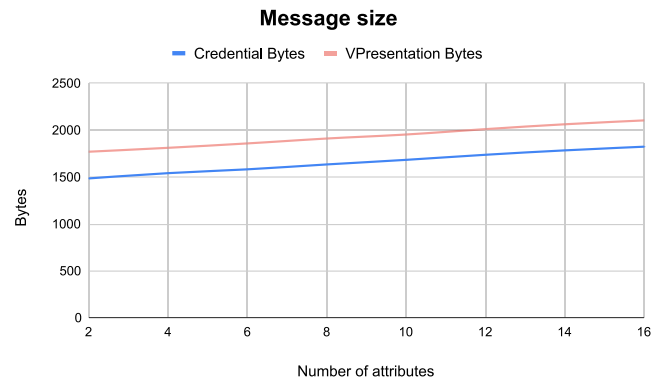


Fig. 15. Metrics of message size with varying number of attributes.

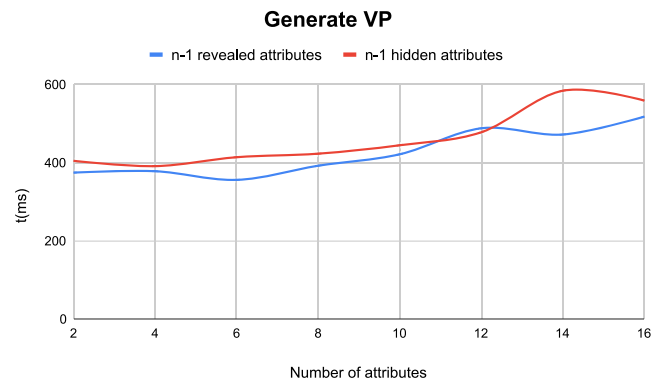
Fig. 16. Comparison of *GenerateVP* execution time for extreme cases in hidden to revealed attributes ratio.

Fig. 14 shows that the results tend to increase linearly as expected, while remaining in the same magnitude order for any number of attributes reasonably needed in most use cases.⁷ Similarly, Fig. 15 shows that message sizes increase linearly with the parameters. The results are again not prohibitive, increasing less than 500 bytes per 15 attributes, as their values are added to the credential structure. This will keep communication overhead in the range of a few kilobytes for most use cases.

To evaluate and visualise the effect of hiding attributes, we show a contrast between the most favourable and unfavourable scenarios. That is, we compare the previous results for $n - 1$ attributes revealed with the same tests, for each test featuring 2, 4, 6, 8, 10, 12, 14, and 16 attributes, $n - 1$ attributes are deliberately hidden. Of course, this only affects the last two processes, which are the only ones executed during authorisation.

As seen in Figs. 16 and 17, the execution time is still increasing linearly in either case. The main difference is that the management of hidden attributes is more expensive than using revealed attributes. This is consistent with the expectations because of the added complexity for the cryptographic operations and the management of the attributes. Nonetheless, the difference is small, keeping the same order of magnitude for the operations, particularly in the case of generating the verifiable presentation. For its part, the difference is more notable in the case of Verifiable Presentation size, as reflected in Fig. 18. While the credential subject includes more (human-readable) identity information when revealing attributes, the size of the credential itself tends to

⁷ For instance, fewer than a hundred attributes, though it would remain efficient even with higher numbers.

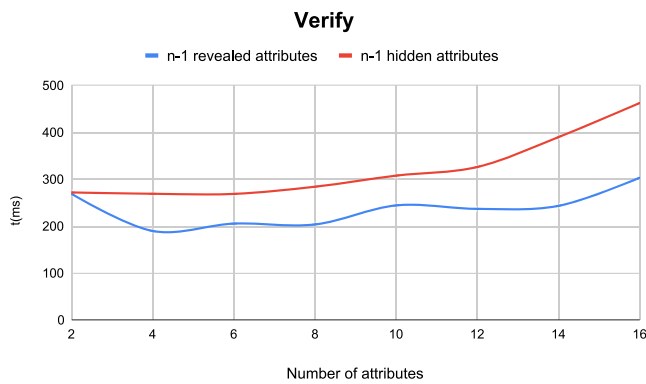


Fig. 17. Comparison of *Verify* execution time for extreme cases in hidden to revealed attributes ratio.

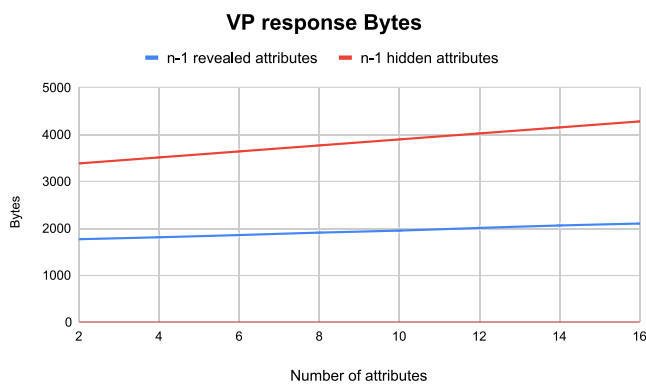


Fig. 18. Comparison of Verifiable Presentation size for extreme cases in hidden to revealed attributes ratio.

be smaller in that case. The reason for this is that the size of the cryptographic proof generated increases linearly with the number of hidden attributes [18]. What is more, the increase is exactly that of an extra field element, which is generally larger (48 plain bytes for the elliptic curve used, BLS-381) than the human-readable representation of the attribute. Even then, the increase in presentation size is still acceptable, as it would lead to a size of ~ 10 KB for a hundred attributes, which is still mostly negligible in the computing continuum use case.

7. Conclusions

This paper has exposed a novel solution to provide decentralised authentication/authorisation across the continuum on distributed architectures, covering different objectives. The proposal provides design, flows and implementation for decentralised enrolment agreement, node enrolment and authorisation during resource acquisition across the continuum. The decentralised Enrolment Agreement allows domains to be considered for being part of the solution. The decentralised node enrolment phase allows domain nodes generating its own identifier and use it for acquiring credentials as well as for publishing its resources information, capabilities and sharing policies. The decentralised authorisation during resource acquisition provides decentralised authorisation features to resource acquisition operations. To this aim, the implementation of the solution relies on DIDs, Verifiable Credentials and DLT technologies, and different methods have been implemented for providing the aforementioned features.

The performance evaluation measures the time spent in the different decentralised authentication and authorisation operations in a

distributed node environment. The results show that the most costly operation is the generation of the node identity which takes less than three seconds. This only occurs once, during node instantiation, so we consider this acceptable. The rest of the operations are in the hundreds of milliseconds mark, taking a total of 786 ms on average for authorisation. Frameworks dedicated to the creation of multi-clusters with sharing and resource acquisition processes, such as Liko within the scope of FLUIDOS, take about 2.5 s without taking into account the authorisation. Thus, our authorisation solution only takes a 31% overhead over those 2.5 s. Therefore, we consider that including a decentralised authorisation such as the one presented in this work does not represent a critical or appreciable overhead to the frameworks, while providing interesting security and privacy features.

CRedit authorship contribution statement

José Manuel Bernabé Murcia: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Conceptualization. **Eduardo Cánovas:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Conceptualization. **Jesús García-Rodríguez:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Investigation, Conceptualization. **Alejandro M. Zarca:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Antonio Skarmeta:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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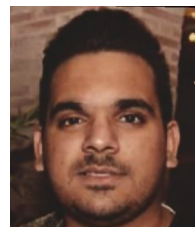
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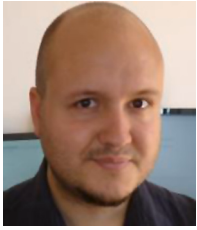
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